20-1-26
ML TUTORIAL - 2

Q) A mobile service Provider wants to classify SMS messages as Spam or Not Spam using a single layer neural network with 5 hidden neurons in layer

x_1	x_2	x_3 (b)	Target (y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

	w_1	w_2	b	
h_1	0.1	0.2	1	$z_1 = w_1x_1 + w_2x_2 + b = 0.4 + 0.4 + 1 = 1.8$
h_2	-0.2	0.3	1	$z_2 = -0.2 + 0 + 1 = 0.8$
h_3	0.4	-0.1	1	$z_3 = \cancel{0.4 + 0.3 + 1 = 1.6}\ 2.4$
h_4	0.3	0.2	1	$z_4 = \cancel{0.6 + 0.2 + 1 = 1.8}\ 2.6$
h_5	0.2	-0.4	1	$z_5 = 1$

$h_1 = \sigma(z_1) = \frac{1}{1+e^{-z}} = 0.$

$\sigma(z_2) = 0.69$

$\sigma(z_3) = 0.$

$\sigma(z_4) = 0.$

$\sigma(z_5) = 0.73$

h_5

$z_{out} = h_1\vartheta_1 + h_2\vartheta_2 + h_3\vartheta_3 + h_4\vartheta_4 + h_5\vartheta_5$

$= 0.86(0.1) + 0.69(0.2) + 0.92(0.3) + 0.93(0$

$+ 0.73(-0.3)$

$= 0.38$

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$$\sigma(z_{out}) = \frac{1}{1+e^{-0.38}}$$

$$= 0.59 \sim 0.60$$

$\hat{y}$ or $y_{pred} = 1$

$\hat{y} = 0 \leftarrow \sigma(z) < 0.5$

or

$1 \leftarrow \sigma(z) \geq 0.5$

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